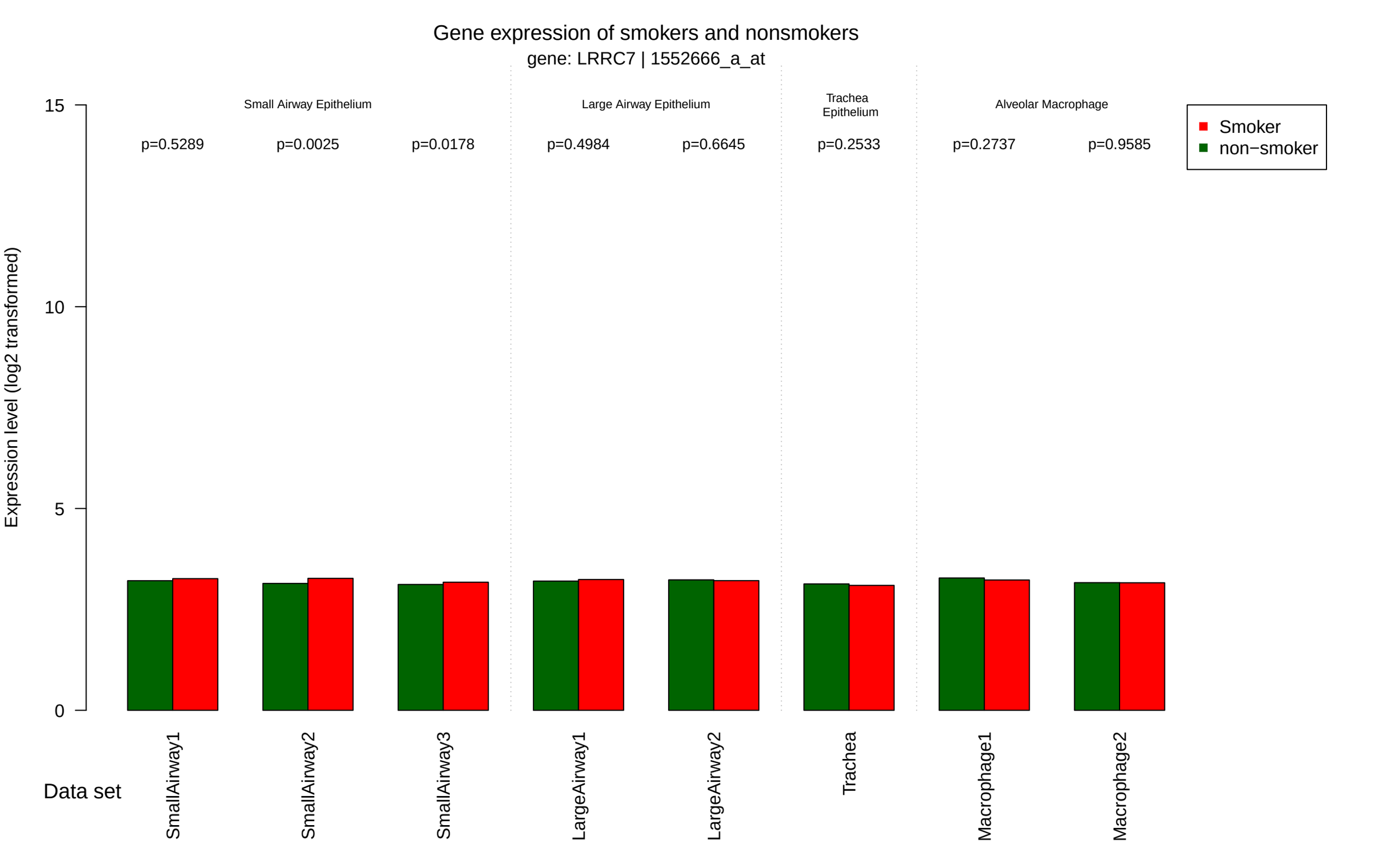
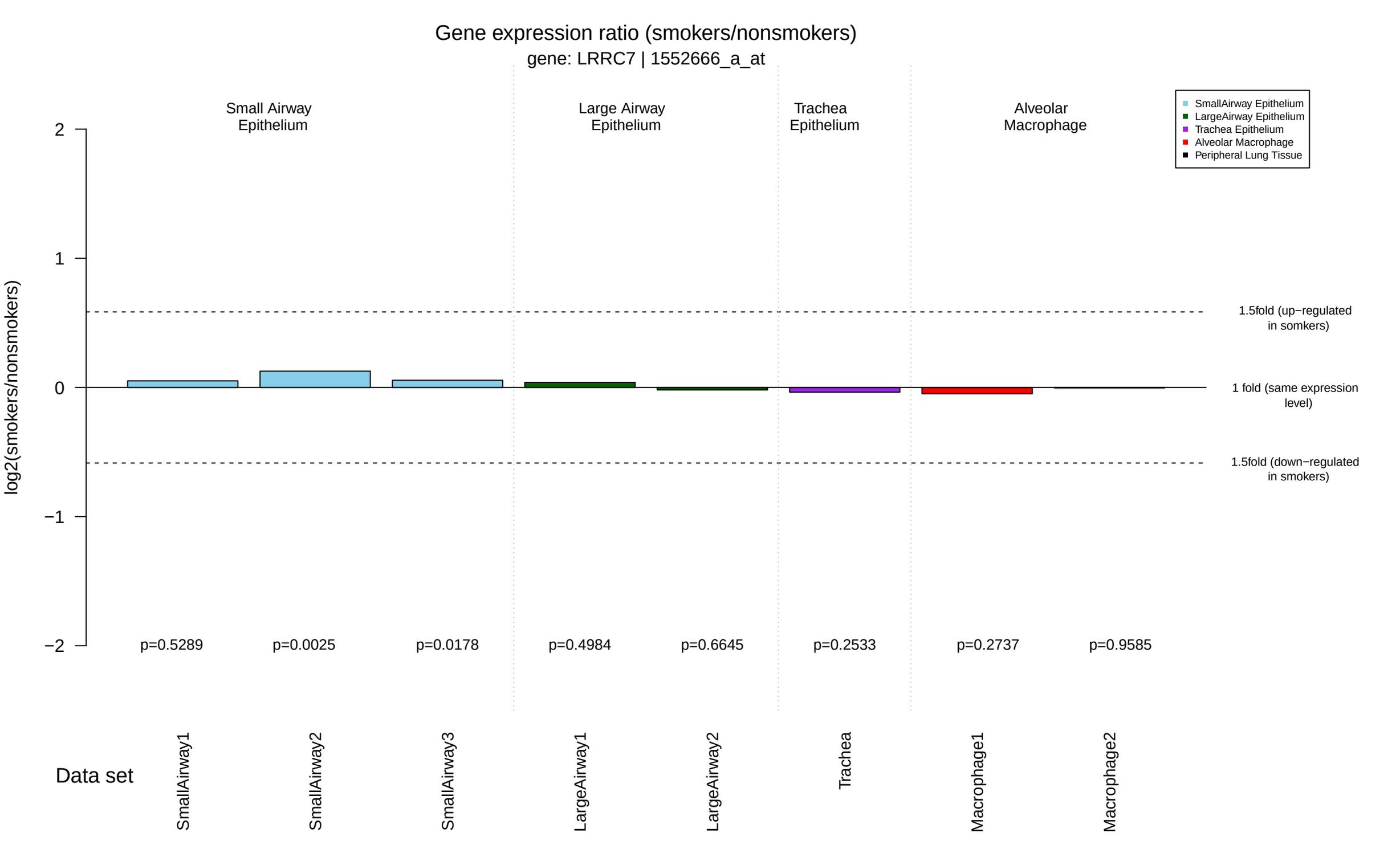
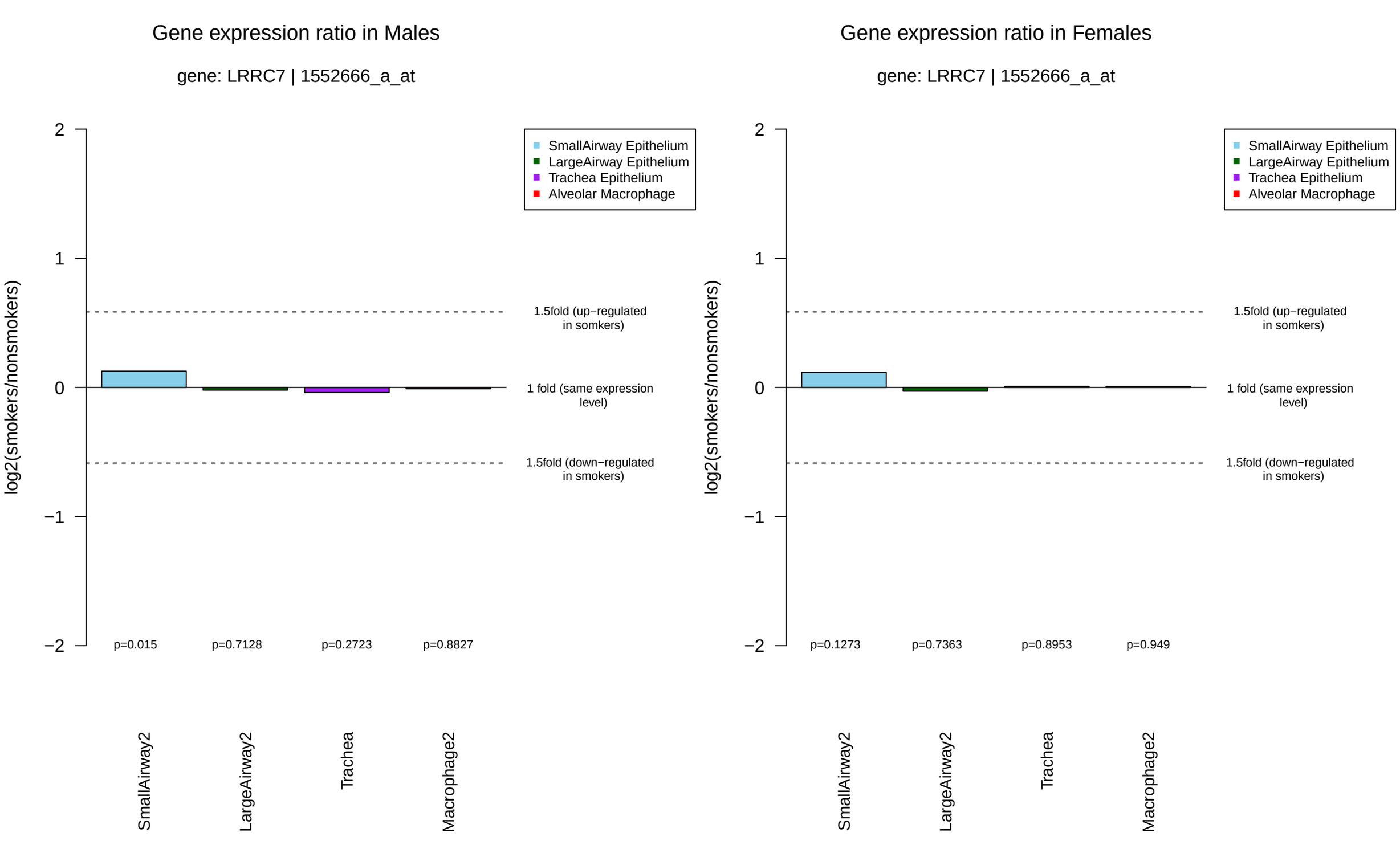
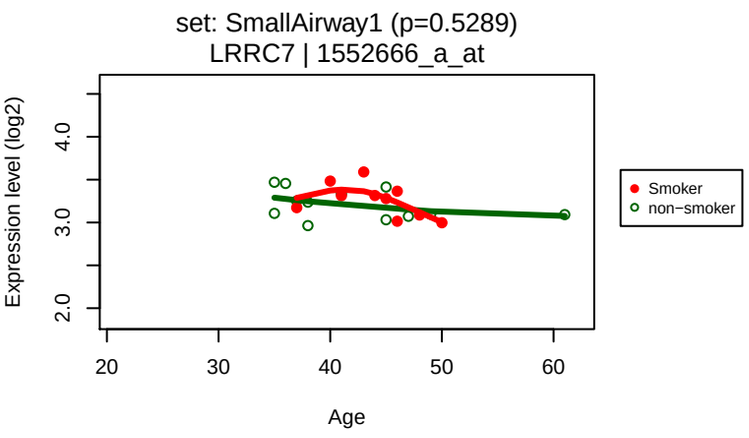
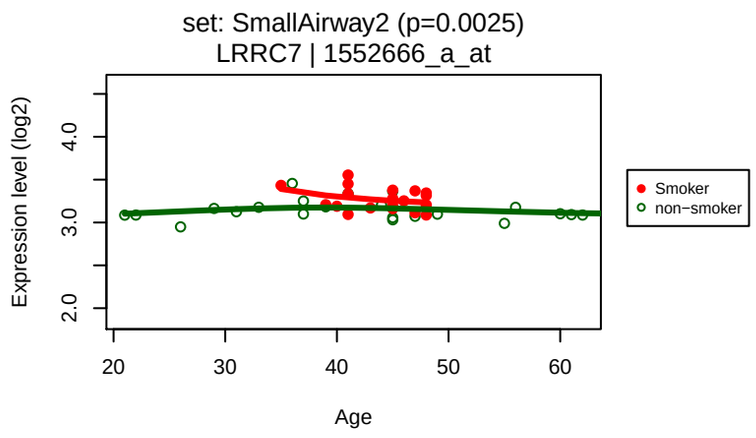
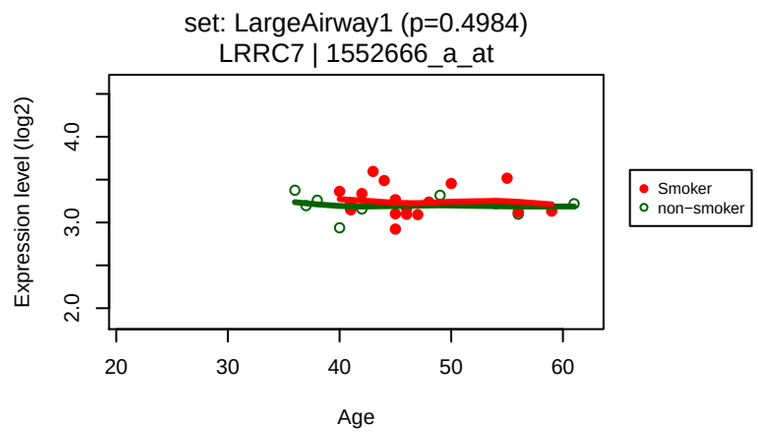

Gene expression of smokers and nonsmokers
gene: LRRC7 | 1552666_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5289
p=0.0025
p=0.0178
p=0.4984
p=0.6645
p=0.2533
p=0.2737
p=0.9585
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

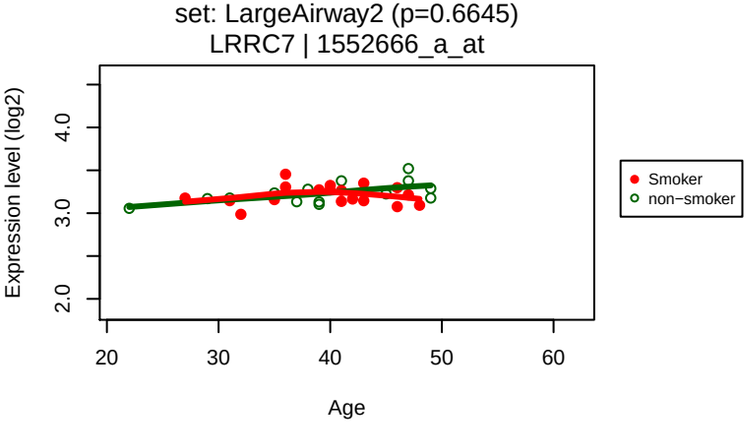

Gene expression ratio (smokers/nonsmokers)
gene: LRRC7 | 1552666_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5289
p=0.0025
p=0.0178
p=0.4984
p=0.6645
p=0.2533
p=0.2737
p=0.9585
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

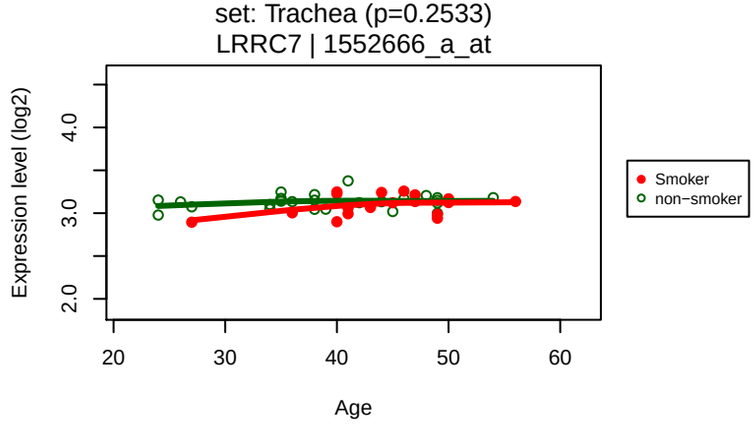

Gene expression ratio in Males
gene: LRRC7 | 1552666_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.015
p=0.7128
p=0.2723
p=0.8827
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LRRC7 | 1552666_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1273
p=0.7363
p=0.8953
p=0.949
SmallAirway2
LargeAirway2
Trachea
Macrophage2

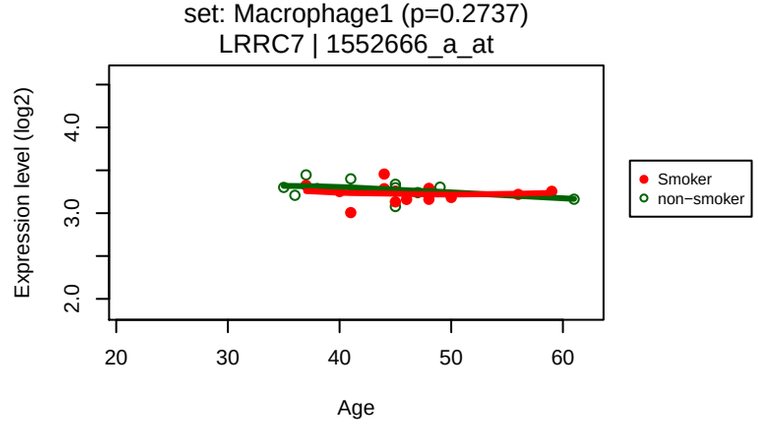

set: SmallAirway1 (p=0.5289)
LRRC7 | 1552666_a_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker

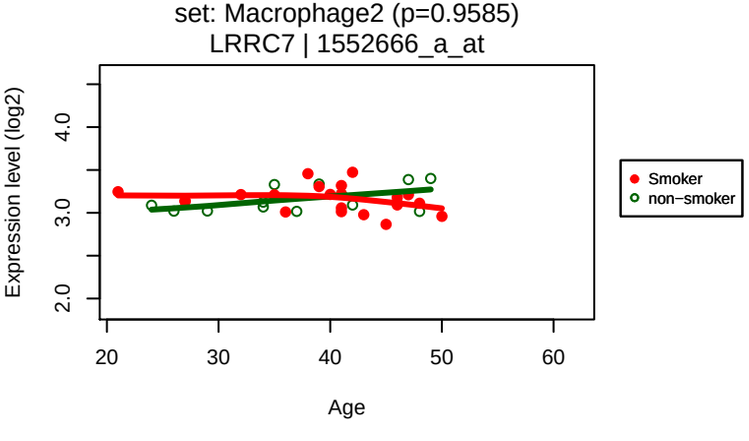

set: SmallAirway2 (p=0.0025)
LRRC7 | 1552666_a_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker

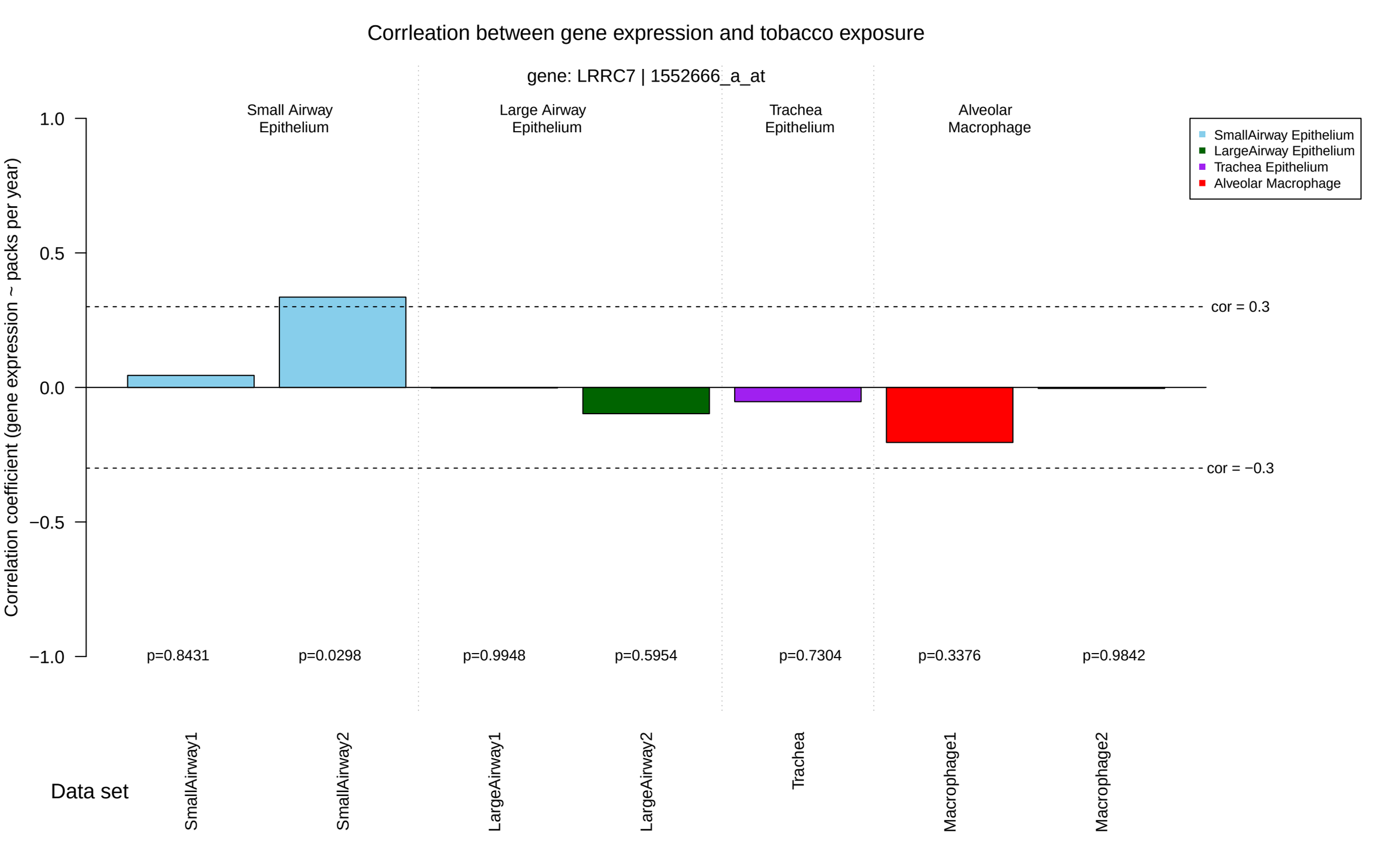

set: LargeAirway1 (p=0.4984)
LRRC7 | 1552666_a_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6645)
LRRC7 | 1552666_a_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.2533)
LRRC7 | 1552666_a_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2737)
LRRC7 | 1552666_a_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9585)
LRRC7 | 1552666_a_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LRRC7 | 1552666_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8431
p=0.0298
p=0.9948
p=0.5954
p=0.7304
p=0.3376
p=0.9842
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2